CHECKING OIL LEVEL

A bayonet type oil gauge at rear of engine is used to determine the quantity of oil in the engine oil pan. This gauge is marked with two designations, "Low" and "Full." Oil should never be allowed to drop below "Low" mark, and no advantage is gained by having oil above "FULL" mark. Dipstick is accessible at rear of cab as illustrated in figure 6.

Daily, or oftener if necessary, check level with engine at normal operating temperature and after engine has stopped for 3 to 5 minutes. Remove dipstick, wipe with clean cloth, replace in crankcase and remove again. Note oil on dipstick and, if necessary, add sufficient oil to bring level up to "FULL" mark. Never run engine if oil is below "LOW" mark.

OIL FILTERS AND STRAINERS

Diesel engines are equipped with at least one oil filter. On some coach engines an oil strainer is also used, and some truck engines use two filters.

OIL FILTER

The oil filters incorporate a replaceable, non-cleanable filtering element through which only a metered portion of the circulated oil passes. These filters remove any impurities which may be detrimental to the engine. Filter element must be discarded and replaced with new element at each oil change; otherwise new oil will become contaminated, and eventually flow of oil through filter will cease resulting in possible serious damage to engine.

MAINTENANCE

Several types of oil filters are used on trucks and coaches, depending upon the particular engine, therefore instructions regarding the replacement of filter element is carried in applicable maintenance manual.

OIL STRAINER

Oil strainer (when used on coach engines) incorporates a cleanable type element through which passes all the oil leaving the oil pump. Strainer removes all particles larger than .005" in diameter.

Strainer element must be thoroughly cleaned at each oil change. If strainer is not cleaned it will eventually become restricted or even clogged; then flow of oil is by-passed around strainer and oil cooler. Under these conditions oil will not be strained or cooled with serious damage to engine resulting.

MAINTENANCE

Refer to applicable maintenance manual covering vehicle being serviced for disassembly, cleaning, and reassembly instructions.

OIL COOLER

Engines are equipped with an oil cooler which not only cools the hot engine, but, by means of the cooling water temperature control, provides a means of rapidly raising the oil temperature during the warm-up period.

The oil cooler is the single-pass, multiple-type, with a corrosion-resistant steel cooling unit contained in a housing. Cooling water drawn through the housing by the water pump completely surrounds the cooling unit. (See fig. 7).

The hot oil enters the cooling unit at the bottom, flows through the inside passages and is discharged at the top into a vertical passage in the cylinder block.